Section A (10 marks)

1. The refrigerant temperature after the expansion device compared to after condenser in the vapor compression refrigeration cycle is
(a) Lower (b) Same (c) Higher (d) None of the above

2. A refrigeration system
(a) Removes heat from a low temperature body and delivers it to a high temperature body
(b) Removes heat from a high temperature body and delivers it to a low temperature body
(c) Rejects energy to low temperature body
(d) None of the above

3. In vapour compression cycle the condition of refrigerant is superheated vapour
(a) after passing through the condenser (b) before passing through the condenser
(c) after passing through the expansion or throttle valve (d) before entering the expansion valve (e) before entering the compressor

4. The bank of tubes at the back of domestic refrigerator are
(a) condenser tubes (b) evaporator tubes (c) refrigerant cooling tubes
(d) capillary tubes (e) throttling device.

5. The higher temperature in vapour compression cycle occurs at
(a) receiver (b) expansion valve (c) evaporator (d) condenser discharge (e) compressor discharge

6. In a refrigeration system, heat absorbed in comparison to heat rejected is
(a) more (b) less (c) same (d) more for small capacity and less for high capacity (e) less for small capacity and more for high capacity.

7. At lower temperatures and pressures, the latent heat of vaporisation of a refrigerant
(a) decreases (b) increases (c) remains same (d) depends on ot factors (e) none of the above.

8. A refrigeration cycle operates between condenser temperature of +27°C and evaporator temperature of **minus 23°C**. The Carnot coefficient of performance of the cycle will be

(a) 0.2 (b) 1.2 (c) 5 (d) 6 (e) 10

9. A domestic refrigerator, set at 2°C, handles on an average a thermal load of 8000 kJ per day. The ambient temperature is 30°C and the COP of the refrigerator is 0.15 times that of a Carnot refrigerator. The daily electricity consumption in kWh is approximately

(a) 1.47 (b) 1.51 (c) 3.28 (d) 2.86

10. One tonne of fish is brought into a cold room and frozen from **+15 °C** to **minus 20 °C** in 18 hours. cp of fish before freezing is **3.2 kJ/kg K and 1.7 kJ/kg K** after freezing; latent heat of fusion (freezing) of fish is **235 kJ/kg** and freezing point of fish is **minus 2 °C**

The product load in **kW** is

(a) 0.49 (b) 0.47 (c) 0.84 (d) 3.63 (e) 4.94

SECTION B (15 marks)

A refrigeration plant operating with a piston compressor and refrigerant R134a shall be built. The required cooling capacity is 25 kW at evaporating temperature of **minus 20°C**. Suction superheat at exit of evaporator shall be 10°C. Condensing temperature shall be +40°C and the condensate sub-cooled in the condenser by 10°C.

i. Draw the cycle very clearly in the attached P-h Chart (**3 marks**)

Assuming that the isentropic and volumetric efficiencies of compressor are 65% and respectively and that the heat loss from the compressor corresponds to 15% of the input power, calculate the following.

ii. Required swept volume (**3 marks**)

iii. Input power of compressor (**3 marks**)

iv. Condenser heat load (**3 marks**)

v. COP. of plant (**3 marks**)

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